

EXP.NO: 5	Scheduling Algorithms-FCFS,SJF,Priority,RR
DATE:13/2/26	

AIM:

To implement CPU Scheduling Algorithms – FCFS, SJF, Priority and Round Robin in Linux.

ALGORITHM:

- FCFS
 1. Execute processes in order of arrival.
 2. Calculate waiting time and turnaround time.
- SJF
 1. Select process with shortest burst time.
 2. Execute until completion.
- Priority
 1. Select process with highest priority.
 2. Execute process accordingly.
- Round Robin
 1. Allocate fixed time quantum.
 2. Execute processes cyclically until completion.

CODE:

```
#include<stdio.h>
int main()
{
    int bt[20], wt[20], tat[20], n, i;
    printf("Enter number of processes: ");
    scanf("%d",&n);
    printf("Enter Burst Time:\n");
    for(i=0;i<n;i++)
        scanf("%d",&bt[i]);
    wt[0]=0;
    for(i=1;i<n;i++)
        wt[i]=wt[i-1]+bt[i-1];
    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
    for(i=0;i<n;i++)
    {
        tat[i]=bt[i]+wt[i];

        printf("P%d\t%d\t\t%d\t\t%d\n",
            i+1,bt[i],wt[i],tat[i]);
    }
    return 0;
}
```

OUTPUT:

```
[hp@localhost ~]$ gcc scheduling.c -o scheduling
[hp@localhost ~]$ ./scheduling
Enter number of processes: 5
Enter Process ID, Burst Time and Priority (PID BT P)
P1 10 3
P2 5 1
P3 8 4
P4 6 2
P5 4 5

===== FCFS =====
Process    AT    BT    CT    TAT    WT
P1         0    10    10    10     0
P2         0    5     15    15    10
P3         0    8     23    23    15
P4         0    6     29    29    23
P5         0    4     33    33    29

Average Waiting Time = 15.40
Average Turnaround Time = 22.00

===== Priority (Non-preemptive) =====
Process    AT    BT    Priority    CT    TAT    WT
P2         0    5        1        5     5     0
P4         0    6        2       11    11     5
P1         0   10        3       21    21    11
P3         0    8        4       29    29    21
P5         0    4        5       33    33    29

Average Waiting Time = 13.20
Average Turnaround Time = 19.80

[hp@localhost ~]$
```

RESULT:

Thus, the Scheduling Algorithms FCFS, SJF, Priority and Round Robin were implemented and executed successfully in Linux.